

SOUMADITYA PAL

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PROFESSIONAL SUMMARY

Full-Stack Platform Engineer with production experience building **real-time** and **distributed systems** using **Next.js, Node.js, PostgreSQL, Redis, and WebSockets**. Experienced in **multi-tenant SaaS architecture, background job processing, state synchronization, and AWS deployments**. Strong focus on **reliability, performance, and system design**.

WORK EXPERIENCE

Emergence Mobility Technologies, Inc

Full Stack Software Engineer | March 2026 – Present | Remote

- Developed high-performance backend services using Node.js and TypeScript, optimizing API response times through targeted architectural and code-level improvements.
- Implemented asynchronous job processing and distributed task queues using BullMQ and Valkey, enabling the system to **handle significantly higher concurrent user load**.
- Designed and built Server-Sent Events (SSE) infrastructure for **real-time data streaming**, eliminating polling overhead and improving perceived client responsiveness.
- Engineered background job pipelines with **retry logic, failure recovery, and state synchronization** using BullMQ, supporting **reliable async workflows under production conditions**.
- Optimized caching and connection management using Valkey, **reducing latency and server load for high-throughput workloads**.
- Applied strong typing and modular architecture patterns in TypeScript to **improve codebase maintainability and reduce runtime errors** across backend services.

Legal Care

Full Stack Developer Intern | Jun 2025 – Sep 2025 | Remote

- Enhanced backend reliability by refactoring Node.js services and patching **critical race conditions** in the distributed authentication flow.
- **Optimized frontend performance** by developing modular React components, translating complex product requirements into responsive, bug-free interfaces.
- Conducted technical code reviews and debugging sessions, improving the maintainability and security of existing backend services.

PROJECTS

FOCUSKARO - High-Availability Task Monitor

Tech: Next.js, BullMQ, Web Workers, Redis

[Live Link](#) | [Github](#)

- Built a **high-availability real-time application** using Next.js, Redis, BullMQ, and Server-Sent Events (SSE).
- Implements **client-side state management** and **failure detection** via heartbeat mechanisms.
- Used **Web Workers** to avoid browser throttling and ensure timer accuracy.
- Designed **background processing and recovery workflows** using Redis queues.
- Designed **lightweight real-time update mechanisms (SSE)** to reduce server overhead and support high-concurrency workloads.

LYNER - Distributed Collaboration Engine

Tech: WebSocket, PostgreSQL, CRDTs, Fastify

[Live Link](#) | [Github](#)

- Developed a **real-time distributed collaboration system** using WebSockets and PostgreSQL.
- Implemented **CRDT-based state synchronization** for eventual consistency.
- Designed **event-sourced architecture** with **append-only logs** for durability and replay.
- Built **low-latency bidirectional communication** using WebSocket protocols.
- Implemented **JWT-based authentication and permission enforcement**.

TECHNICAL SKILLS

Backend: Node.js, Concurrency, Async Processing, State Management, WebSockets, SSE, Redis, BullMQ, Fastify, CRDTs, JWT-based authentication.

Databases: PostgreSQL (schema design, indexing, transactions), Redis (caching, queues), Valkey.

Frontend: Next.js, React.js, Web Workers.

Languages: TypeScript, JavaScript (ES6+).

Tools: AWS, Deployment, CI/CD.

EDUCATION

NSHM Knowledge Campus, Durgapur

Bachelor of Technology, Computer Science and Engineering | 2022-2026